

```
install.packages("googledrive")
library(googledrive)
```

```
drive_auth()
```

```
drive_download("mortgage.dat", path = "mortgage.dat", overwrite = TRUE)
```

```
data <- read.table("mortgage.dat", header = TRUE, sep = "")
```

⇒ Installing package into '/usr/local/lib/R/site-library' (as 'lib' is unspecified)

The **googledrive** package is requesting access to your Google account.
Enter '1' to start a new auth process or select a pre-authorized account.

1: Send me to the browser for a new auth process.

2: maaaanideep@gmail.com

Selection: 1

Please point your browser to the following url:

https://accounts.google.com/o/oauth2/v2/auth?client_id=603366585132-frj1ouoa3s2o

Enter authorization code: eyJjb2RlIjoicNC8wQWVhblMwWjlyay0xeWtYNWlYUkVLbk9KU1NrY1

File downloaded:

- mortgage.dat <id: 10UQfuaRS64-sR2FI0i00FDVZhUFD1WZg>

Saved locally as:

- [mortgage.dat](#)

part a

```
data <- read.table("mortgage.dat", header = TRUE, sep = "")
```

```
print(head(data))
```

```
print(names(data))
```

⇒

	X1976	X06	X01	X8.85
1	1976	7	1	8.93
2	1976	8	1	9.00
3	1976	9	1	8.98
4	1976	10	1	8.93
5	1976	11	1	8.81
6	1976	12	1	8.79

[1] "X1976" "X06" "X01" "X8.85"

```
colnames(data) <- c("Year", "Month", "Day", "Rate")
```

```
data$Date <- as.Date(with(data, paste(Year, Month, Day, sep = "-")))

time_series <- ts(data$Rate, start = c(data$Year[1], data$Month[1]), frequency = 12)

plot(time_series, main = "30-Year Mortgage Rate ",
      ylab = "Rate (%)", col = "black", lwd = 2)
```



```
install.packages("tseries")
```



```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
also installing the dependencies 'xts', 'TTR', 'quadprog', 'zoo', 'quantmod'
```

```
Registered S3 method overwritten by 'quantmod':
  method      from
as.zoo.data.frame zoo
```

```
library(tseries)
adf_result <- adf.test(time_series)
cat("ADF Test for Original Series:\n")
cat("Dickey-Fuller Statistic:", adf_result$statistic, "\n")
cat("p-value:", adf_result$p.value, "\n")
```

```
➦ ADF Test for Original Series:  
Dickey-Fuller Statistic: -3.075113  
p-value: 0.1231679
```

With a p-value of 0.1231679, the null hypothesis of the ADF test cannot be rejected. This indicates the time series is non-stationary and needs transformation.

part b

```
log_time_series <- log(time_series)  
  
plot(log_time_series, main = "Log-Transformed 30-Year Mortgage Rate",  
     ylab = "Log(Rate)", col = "black", lwd = 2)
```



It shows a clear downward trend, particularly from 1980s which indicates that series is still not stationary, The transformation reduced variability but it needs further differencing.

part c

```
diff_log_time_series <- diff(log_time_series)  
  
plot(diff_log_time_series, main = "Differenced Log-Transformed Series",
```

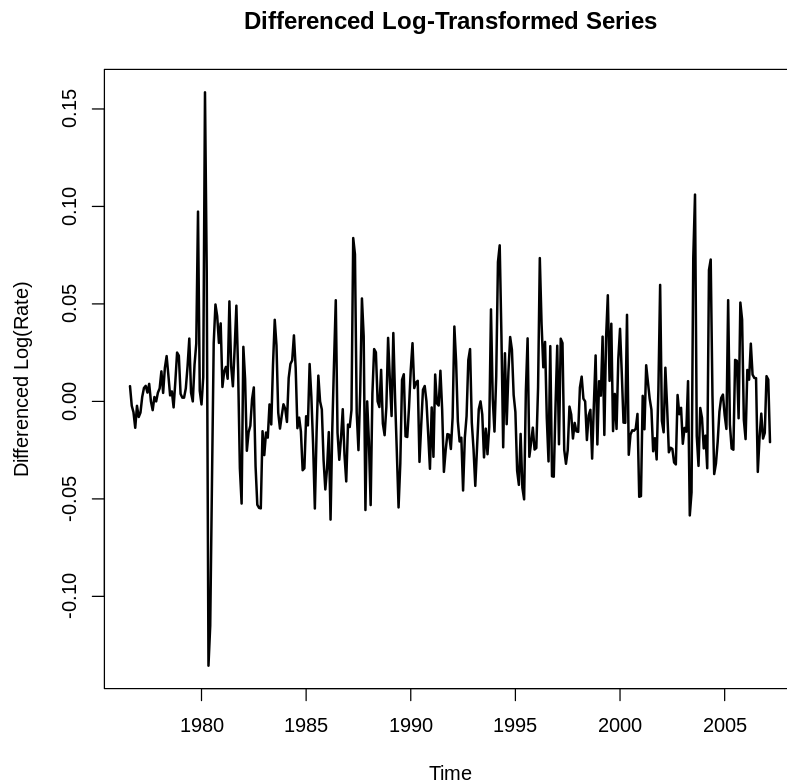
```
ylab = "Differenced Log(Rate)", col = "black", lwd = 2)
```

```
adf_diff_result <- adf.test(diff_log_time_series)
print(adf_diff_result)
```

Warning message in `adf.test(diff_log_time_series)`:
"p-value smaller than printed p-value"

Augmented Dickey-Fuller Test

```
data: diff_log_time_series
Dickey-Fuller = -6.2366, Lag order = 7, p-value = 0.01
alternative hypothesis: stationary
```



This now fluctuates around a constant mean, has no trend or seasonality. ADF test's p-value suggests that the null-hypothesis of non-stationarity can be rejected. spikes around 1980 may be due to other external reasons.

part d

```
install.packages("TSA")
```

Installing package into `'/usr/local/lib/R/site-library'`
(as `'lib'` is unspecified)

also installing the dependencies `'leaps'`, `'locfit'`

```
library(TSA)
acf(diff_log_time_series, main = "ACF of Differenced Log Series")
pacf(diff_log_time_series, main = "PACF of Differenced Log Series")
eacf(diff_log_time_series)
```



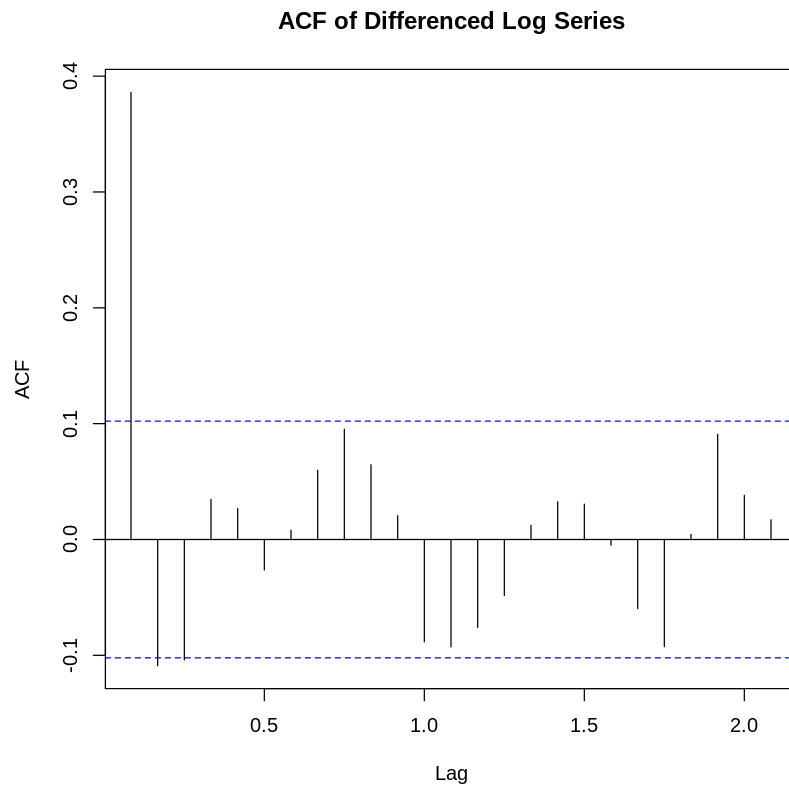
Attaching package: 'TSA'

The following objects are masked from 'package:stats':

acf, arima

The following object is masked from 'package:utils':

tar

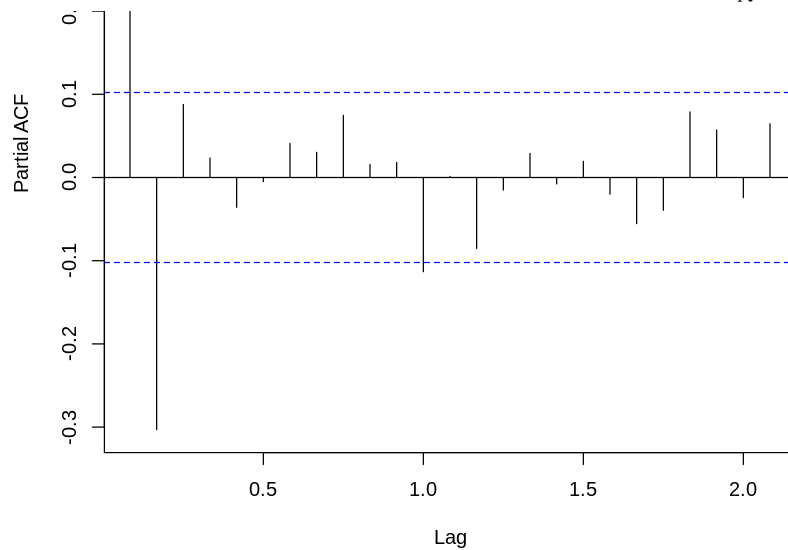


AR/MA

	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	x	x	0	0	0	0	0	0	0	0	0	0	0	0
1	x	x	0	x	0	0	0	0	0	0	0	0	0	0
2	x	x	0	0	0	0	0	0	0	0	0	0	0	0
3	x	x	0	0	0	0	0	0	0	0	0	0	0	0
4	x	x	0	0	0	0	0	0	0	0	0	0	0	0
5	x	x	x	0	0	0	0	0	0	0	0	0	0	0
6	x	x	x	0	0	0	0	0	0	0	0	0	0	0
7	x	x	x	x	0	0	0	0	0	0	0	0	0	0

PACF of Differenced Log Series





The ARMA(2,1) model is suggested by residual analysis. we can also try ARMA(3,1) and ARMA(2,2).

```
m0 <- arima(diff_log_time_series, order = c(2, 0, 1))
```

```
print(m0)
```

```
cat("AR(1) Coefficient:", m0$coef["ar1"], "Standard Error:", sqrt(diag(m0$var.coef))
cat("AR(2) Coefficient:", m0$coef["ar2"], "Standard Error:", sqrt(diag(m0$var.coef))
cat("MA(1) Coefficient:", m0$coef["ma1"], "Standard Error:", sqrt(diag(m0$var.coef))
```



Call:

```
arima(x = diff_log_time_series, order = c(2, 0, 1))
```

Coefficients:

	ar1	ar2	ma1	intercept
	0.3073	-0.2294	0.2182	-0.0010
s.e.	0.1373	0.0760	0.1378	0.0018

sigma^2 estimated as 0.0006865: log likelihood = 817.87, aic = -1627.75

AR(1) Coefficient: 0.3072549 Standard Error: 0.1373347

AR(2) Coefficient: -0.2294071 Standard Error: 0.07596327

MA(1) Coefficient: 0.218246 Standard Error: 0.137789

```
m0 <- arima(diff_log_time_series, order = c(2, 0, 1))
```

```
print(m0)
```

```
plot(m0$residuals, main = "Residuals of ARMA(2,1)", ylab = "Residuals", col = "black"
abline(h = 0, col = "green", lty = 2)
```

```
acf(m0$residuals, main = "ACF of Residuals")
```

```
pacf(m0$residuals, main = "PACF of Residuals")
```

```
Box.test(m0$residuals, lag = 12, type = "Ljung-Box", fitdf = 3)  
cat("AIC of the ARMA(2,1) model:", m0$aic, "\n")
```




Call:

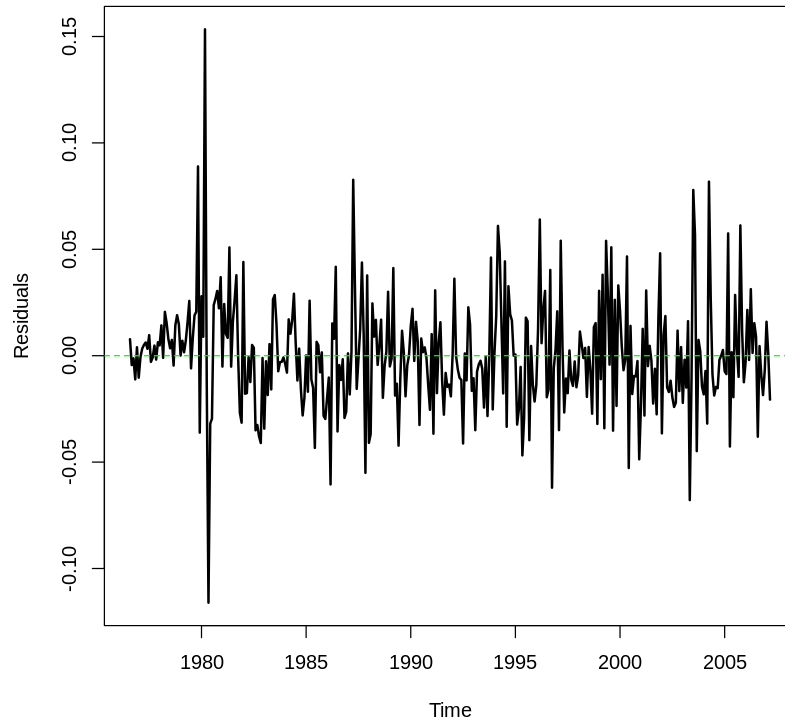
```
arma(x = diff_log_time_series, order = c(2, 0, 1))
```

Coefficients:

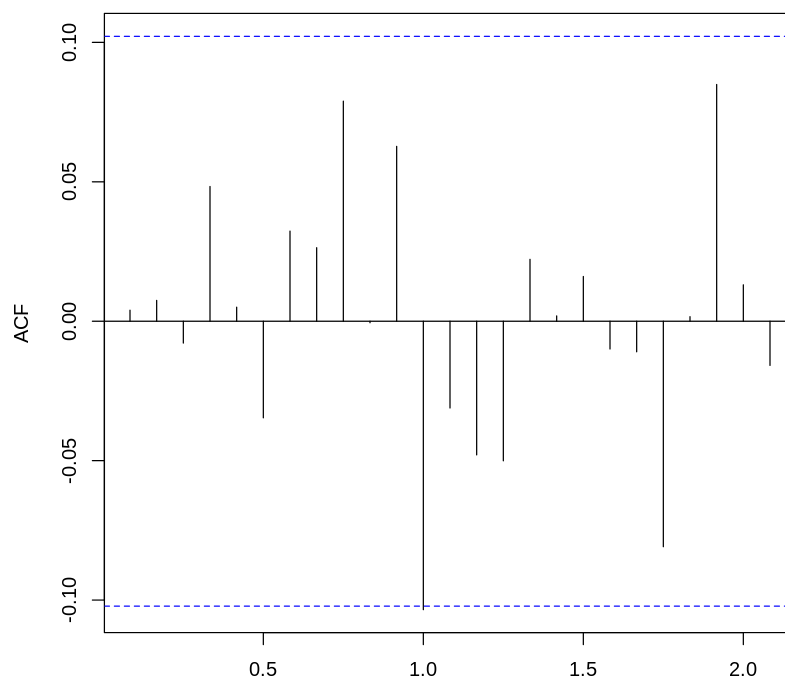
	ar1	ar2	ma1	intercept
	0.3073	-0.2294	0.2182	-0.0010
s.e.	0.1373	0.0760	0.1378	0.0018

sigma^2 estimated as 0.0006865: log likelihood = 817.87, aic = -1627.75

Residuals of ARMA(2,1)



ACF of Residuals



Lag

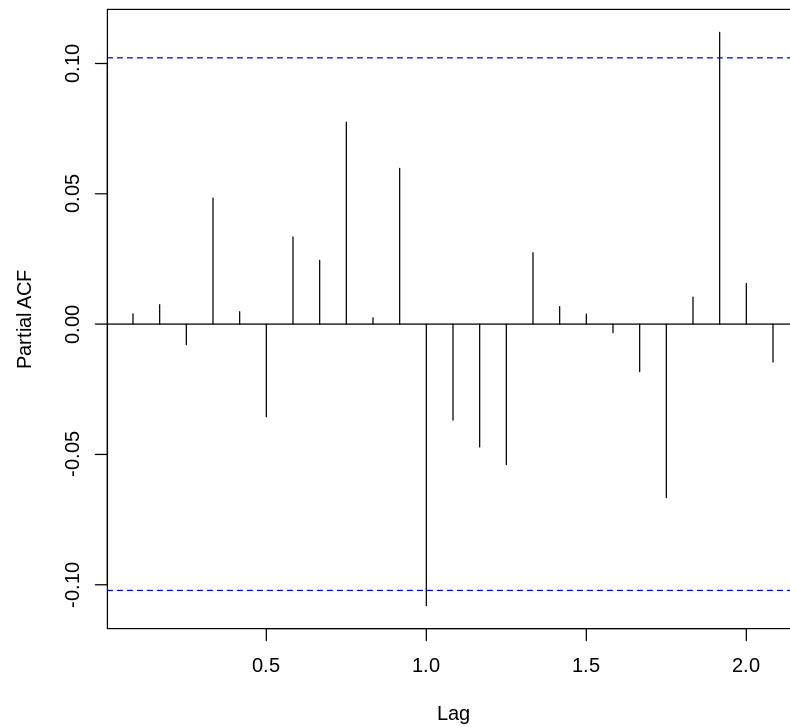
Box-Ljung test

data: m0\$residuals

X-squared = 9.9947, df = 9, p-value = 0.3509

AIC of the ARMA(2,1) model: -1627.749

PACF of Residuals



The residuals fluctuate randomly around zero with no trend or seasonality, confirming the ARMA(2,1) model effectively captures the time series dynamics. The ACF, PACF, and EACF validate uncorrelated residuals, while the Box-Ljung test ($p > 0.05$) confirms white noise. A low AIC indicates a strong model fit.

```
cat("AIC:", m0$aic, "\n")
```

```
➡ AIC: -1625.749
```

```
calculate_rolling_forecast_error <- function(series, model_order, window_size) {
  T <- length(series)
  errors <- c()
  for (t in (T - window_size):(T - 1)) {
    train_data <- series[1:t]
    test_value <- series[t + 1]
    fitted_model <- arima(train_data, order = model_order)
    forecast_value <- predict(fitted_model, n.ahead = 1)$pred
    error <- (forecast_value - test_value)^2
    errors <- c(errors, error)
  }
  return(mean(errors))
}
```

```
avg_squared_error <- calculate_rolling_forecast_error(
  diff_log_time_series, c(2, 0, 1), 30
)
cat("Average Squared Error:", avg_squared_error, "\n")
```

```
➡ Average Squared Error: 0.0005065388
```

```
# ARMA(3,1)
arma31 <- arima(diff_log_time_series, order = c(3, 0, 1))
print(arma31)
```

```
plot(arma31$residuals, type = "l", main = "Residuals of ARMA(3,1)", ylab = "Residual
abline(h = 0, col = "green", lty = 2)
```

```
acf(arma31$residuals, main = "ACF of Residuals (ARMA(3,1))")
pacf(arma31$residuals, main = "PACF of Residuals (ARMA(3,1))")
```

```
box_ljung_31 <- Box.test(arma31$residuals, lag = 12, type = "Ljung-Box", fitdf = 4)
print(box_ljung_31)
```

```
cat("AIC of ARMA(3,1):", arma31$aic, "\n")
```



Call:

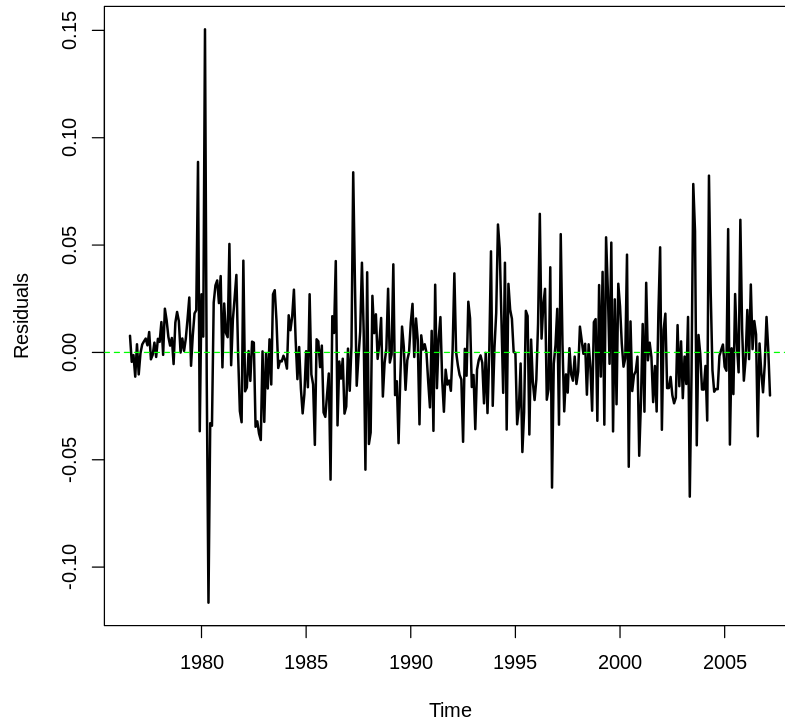
```
arma(x = diff_log_time_series, order = c(3, 0, 1))
```

Coefficients:

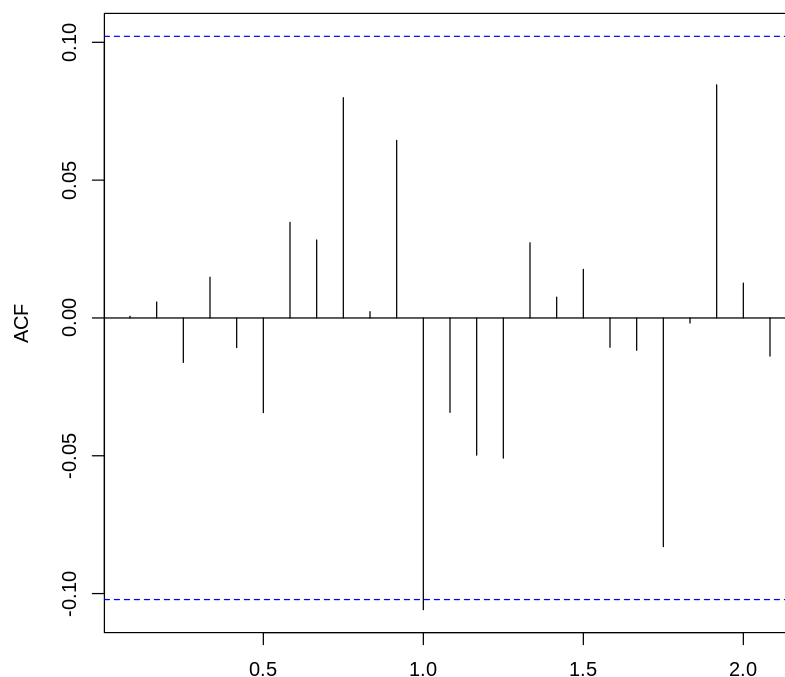
	ar1	ar2	ar3	ma1	intercept
	0.6791	-0.4227	0.1337	-0.1500	-0.0010
s.e.	0.4335	0.2224	0.1351	0.4359	0.0019

sigma² estimated as 0.0006852: log likelihood = 818.23, aic = -1626.45

Residuals of ARMA(3,1)



ACF of Residuals (ARMA(3,1))



Lag

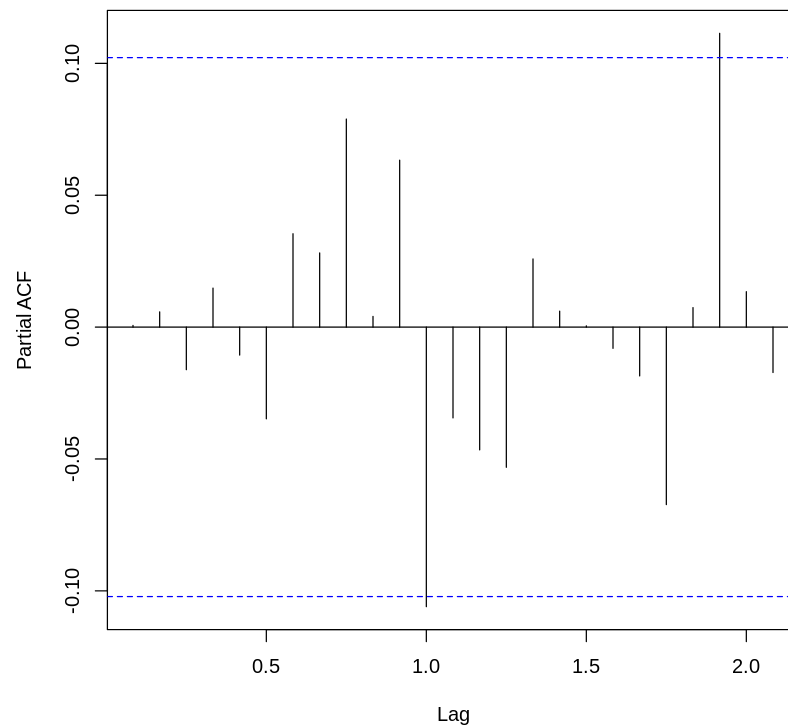
Box-Ljung test

data: arma31\$residuals

X-squared = 9.7254, df = 8, p-value = 0.2848

AIC of ARMA(3,1): -1626.452

PACF of Residuals (ARMA(3,1))



The residuals for the ARMA(3,1) fluctuate around zero with no trend or seasonality. The ACF and PACF plots show no significant autocorrelations, confirming uncorrelated residuals. The Box-Ljung test ($p > 0.05$) supports the residuals being white noise. A low AIC value indicates a good fit.

```
arma22 <- arima(diff_log_time_series, order = c(2, 0, 2))
print(arma22)

plot(arma22$residuals, type = "l", main = "Residuals of ARMA(2,2)", ylab = "Residual",
     abline(h = 0, col = "blue", lty = 2))

acf(arma22$residuals, main = "ACF of Residuals (ARMA(2,2))")
pacf(arma22$residuals, main = "PACF of Residuals (ARMA(2,2))")

box_ljung_22 <- Box.test(arma22$residuals, lag = 12, type = "Ljung-Box", fitdf = 4)
print(box_ljung_22)

cat("AIC of ARMA(2,2):", arma22$aic, "\n")
```



Call:

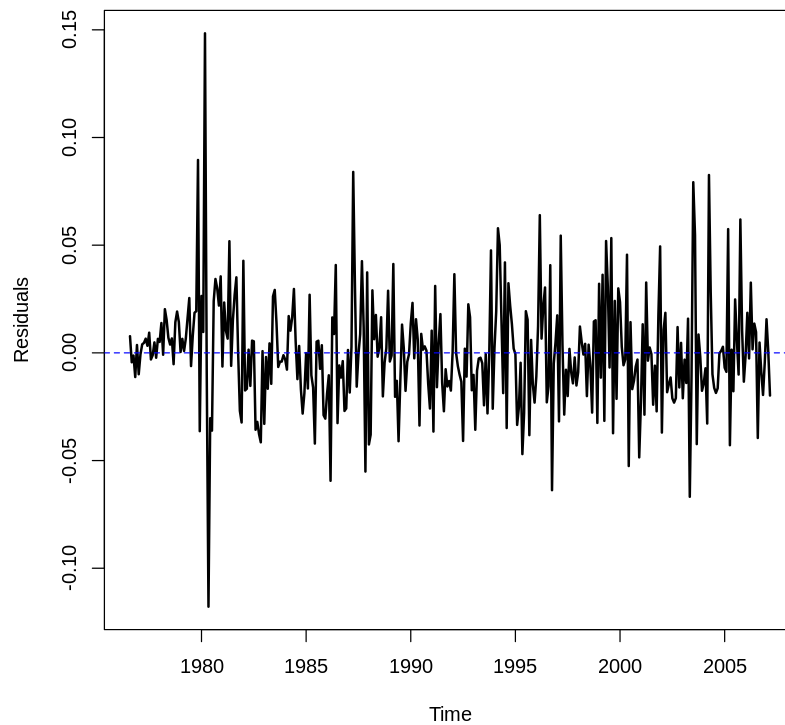
```
arma(x = diff_log_time_series, order = c(2, 0, 2))
```

Coefficients:

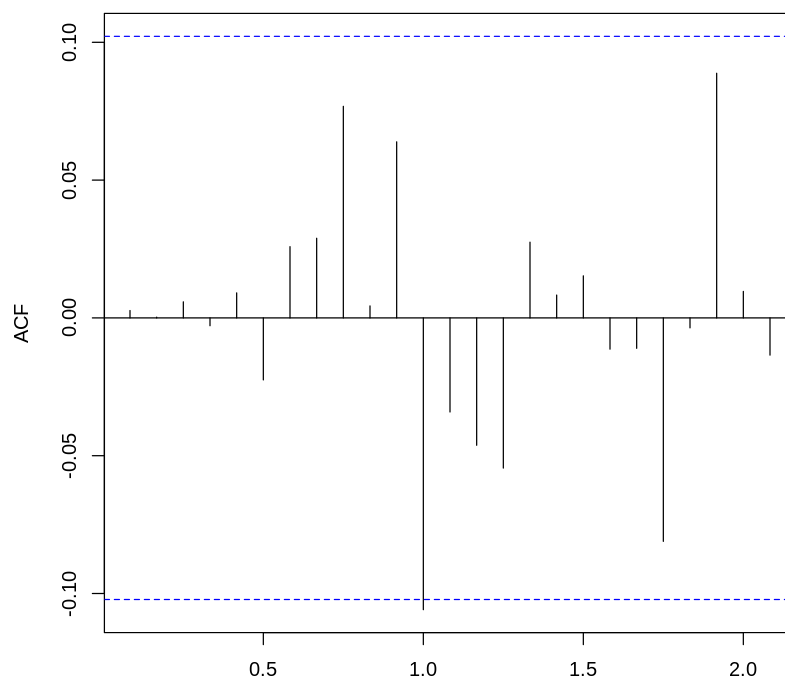
	ar1	ar2	ma1	ma2	intercept
	0.0122	-0.2866	0.5149	0.2221	-0.0010
s.e.	0.2795	0.0944	0.2814	0.1724	0.0019

sigma² estimated as 0.0006843: log likelihood = 818.47, aic = -1626.94

Residuals of ARMA(2,2)



ACF of Residuals (ARMA(2,2))



Lag

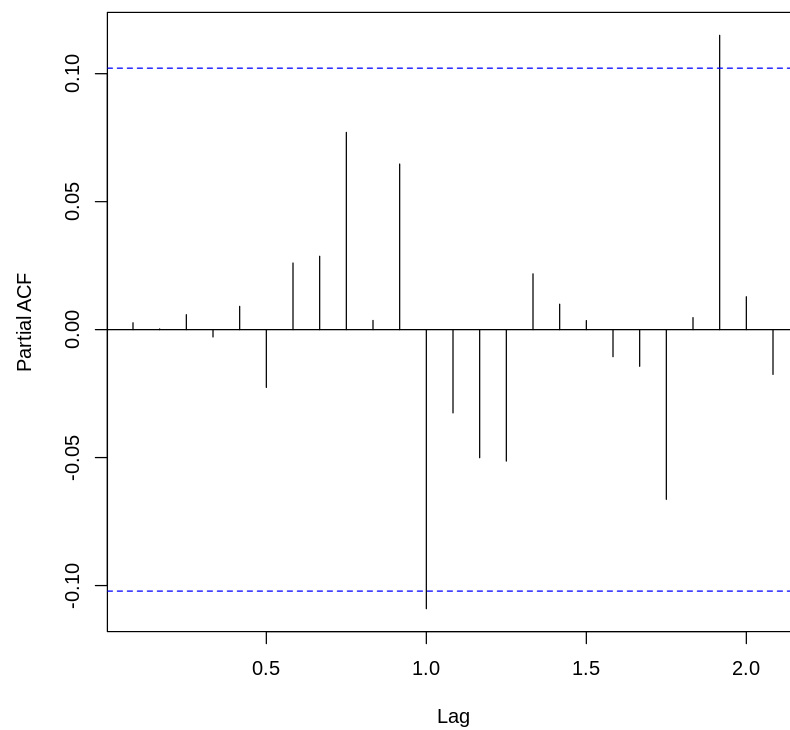
Box-Ljung test

data: arma22\$residuals

X-squared = 8.8922, df = 8, p-value = 0.3515

AIC of ARMA(2,2): -1626.937

PACF of Residuals (ARMA(2,2))



The ARMA(2,2) residuals also show random fluctuations around zero with no discernible trend, indicating the model captures the time series well. ACF and PACF plots exhibit no significant spikes, confirming uncorrelated residuals, while the Box-Ljung test ($p > 0.05$) verifies the residuals are white noise. Although the AIC value is low, the slight increase in complexity compared to ARMA(2,1) may not justify any marginal improvements.

```
cat("AIC Comparison:\n")
cat("ARMA(2,1):", m0$aic, "\n")
cat("ARMA(3,1):", arma31$aic, "\n")
cat("ARMA(2,2):", arma22$aic, "\n")
```

```
⇒ AIC Comparison:
   ARMA(2,1): -1627.749
   ARMA(3,1): -1626.452
   ARMA(2,2): -1626.937
```

Based on the AIC comparison, the ARMA(2,1) model is the best choice as it has the lowest AIC value (-1627.749), showing it provides the best fit with the least complexity. The ARMA(3,1) (-1626.452) and ARMA(2,2) (-1626.937) models have slightly higher AIC values, meaning they do not significantly improve the fit while adding more complexity. The residual diagnostics for ARMA(2,1), including ACF, PACF, and the Box-Ljung test ($p > 0.05$), confirm that the residuals are